EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Getting a correct output once guarantees an algorithm is bug-free.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

DO NOW — RECALL

Paper 2 examiner-style feedback

Name: _____ Class: _____ Date: _____

1. Why must recursive calls progress?

2. What does a midpoint calculate?

3. What happens when low>high?

4. C# [2,4,6] with a[1]=a[0]+a[2]. Final array?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

To reach the base case.

2. Model answer

The middle index of remaining interval.

3. Model answer

No candidates remain; not found.

4. Model answer

[2,8,6].

By the end, you can…

- Identify whether the error lies in problem reading, algorithm state or pseudocode.
- Explain every bound update, recursive return or data-structure operation.
- Retest the method with a different input to demonstrate transfer.

Vocabulary — know these terms

mark scheme — Criteria for awarding marks.

termination — Algorithm eventually stops.

auxiliary space — Additional storage beyond the input.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Correct: 'merge sort has O(1) auxiliary space because the temp array is short-lived'. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

An algorithm's recursion has no base case. Explain fault and give two repairs. [6]

Question 2

Trace binary search for 9 in [1,3,5,7,9], recording every low, high and midpoint. [6]

NEW LEARNING

Read, annotate and explain

Understand

Identify whether the error lies in problem reading, algorithm state or pseudocode.

Apply

Explain every bound update, recursive return or data-structure operation.

Evaluate

Retest the method with a different input to demonstrate transfer.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Repair a SumTo recursion missing a base case.

Explain the result

The n==0 base case stops recursion and returns zero.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve